Bounding the extension power of the Hodge line bundle on period domains constructed from quaternion algebras

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On a period domain constructed from $r \in \{2, 3\}$ copies of a rational quaternion algebra $A_{\alpha,\beta}$ ($\alpha, \beta \in \mathbb{Q}_{<0}$), we uniformly bound over all boundary components the extension character χ_∞ under Satake-Baily-Borel compactification, for all α, β and every lattice $V_\mathbb{Z} \subset A_{\alpha,\beta}^{\oplus r}$, analogously to the methods of Green, Griffiths, and Robles in [GGR]. For $r = 2$, $\chi_\infty = 1$, so the Hodge line bundle on the period domain extends to the compactification on the nose. For $r = 3$, the extension character is controlled by α alone; if d is the positive squarefree part of α, then the power of the Hodge line bundle that is needed for it to extend divides 2 when $d = 1$, and it divides 3 when $d = 3$. Otherwise, the Hodge line bundle extends on the nose. We also exhibit constructions that attain these bounds.

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2 Quaternions, conventions, construction, and the extension character

We will take $\alpha, \beta \in \mathbb{Q}_{<0}$ to be arbitrary negative rationals, and we take $r \in \{2, 3\}$.

We define the quaternion algebra with abstract generators $\mathbf{I}, \mathbf{J}, \mathbf{K}$ satisfying the relations

$$\mathbf{I}^2 = \alpha, \mathbf{J}^2 = \beta, \mathbf{K} = \mathbf{IJ} = -\mathbf{JI}$$

which forces

$$\mathbf{K}^2 = -\alpha\beta, \mathbf{IK} = \alpha\mathbf{J} = -\mathbf{KI}, \mathbf{KJ} = \beta\mathbf{I} = -\mathbf{JK}.$$

This gives the quaternion algebra over $\mathbb{Q}$:

$$A_{\mathbb{Q}} = \text{span}_{\mathbb{Q}}\{1, \mathbf{I}, \mathbf{J}, \mathbf{K}\} \cong \mathbb{Q}^4. \tag{2.0.1}$$

We can thus conflate the element $(w, x, y, z) \in \mathbb{Q}^4$ with the element $w + x\mathbf{I} + y\mathbf{J} + z\mathbf{K} \in A$. This yields the standard product

$$\begin{aligned}(w,x,y,z) \times (w',x',y',z') \;=\; & (ww' + xx'\alpha + yy'\beta - zz'\alpha\beta\,, \\ & wx' + xw' + zy'\beta - yz'\beta\,, \\ & wy' + yw' + xz'\alpha - zx'\alpha\,, \\ & wz' + zw' + xy' - yx')\,.\end{aligned}$$

The conjugate of $a = (w, x, y, z)$ is $\overline{a} = (w, -x, -y, -z)$. If $a = -\overline{a}$ (i.e. $w = 0$) we call a a *pure* quaternion. We also have an inner product

$$(w,x,y,z) \cdot (w',x',y',z') \;=\; w\,w' \;-\; \alpha\,x\,x' \;-\; \beta\,y\,y' \;+\; \alpha\beta\,z\,z'\,.$$

which is a symmetric bilinear form Q on our algebra, with matrix

$$Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -\alpha & 0 & 0 \\ 0 & 0 & -\beta & 0 \\ 0 & 0 & 0 & \alpha\beta \end{bmatrix}$$

We now list some useful properties of quaternions:

Lemma 2.1 (Properties of quaternions)**.** *For $\alpha, \beta < 0$, the algebras $A_{\mathbb{Q}}$ and $A_{\mathbb{R}} = A_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R}$ are division algebras. However, note that $A_{\mathbb{C}} = A_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C} \cong M_2(\mathbb{C})$ is not a division algebra. For a quaternion $q = (w, x, y, z) = w + x\boldsymbol{I} + y\boldsymbol{J} + z\boldsymbol{K} \in A_{\mathbb{Q}}$ we have*

1. *The norm of q is $\text{Norm}(q) = w^2 - \alpha x^2 - \beta y^2 + \alpha\beta z^2 = q\overline{q} = \overline{q}q > 0$ for any nonzero q*
2. *The trace of q is $\text{Tr}(q) = 2w = q + \overline{q}$*
3. *The reduced characteristic polynomial of q is*

   $$P_q(x) = x^2 - \text{Tr}(q)x + \text{Norm(q)}$$

   Note that any quaternion q satisfies its own characteristic polynomial P_q: we have $q^2 - \text{Tr}(q)q + \text{Norm}(q) = 0$. On the 4-dimensional spaces $A_{\mathbb{Q}}$ and $A_{\mathbb{C}}$, the characteristic polynomial of left multiplication by q is P_q^2.
4. *$\overline{a \times b} = \overline{b} \times \overline{a}$ for all $a, b \in A_{\mathbb{Q}}$;*
5. *If q is pure then $q^2 = -\text{Norm}(q) \leq 0$, with equality iff $q = 0$.*

Proof. Detailed explanations can be found in [Voi21]. (1), (2), and (4) come from expanding the definitions above. The norm is positive because every term has positive coefficients. Since the norm is positive and multiplicative $A_{\mathbb{Q}}$ is thus a division algebra; this is because if $q_1 q_2 = 0$ then $\text{Norm(q}_1\text{)Norm(q}_2\text{)} = 0$ and the only quaternion with zero norm is zero. One also can show that the inverse of q is $\frac{\overline{q}}{\text{Norm}(q)}$. For (3) the polynomial and the left multiplication statement can be found in [Voi21, Ch.2, Ch.3], and by Cayley-Hamilton q must satisfy its own characteristic polynomial so $P_q(q) = 0$. This fact can also be seen by writing $P_q(q) = q^2 - (q + \overline{q})q + q\overline{q} \cdot 1 = q^2 - q^2 - \overline{q}q + \overline{q}q = 0$. For (5), for some pure q we have $\overline{q} = -q$ so $\text{Norm}(q) = q\overline{q} = -q^2$. □

Lemma 2.2 (Complex determinants of left multiplications)**.** *Let $A_{\mathbb{C}} = A_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C} \simeq M_2(\mathbb{C})$ and let $L_u : A_{\mathbb{C}} \to A_{\mathbb{C}}$ denote left multiplication by $u \in A_{\mathbb{Q}}^{\times}$. Then the eigenvalues of L_u are the two roots λ, λ' of the reduced characteristic polynomial P_u, each with multiplicity two, and*

$$\det_{\mathbb{C}}(L_u) \;=\; \mathrm{Norm}(u)^2$$

Moreover, if $\mathrm{Norm}(u) = 1$*, then $\lambda' = \overline{\lambda} = \lambda^{-1}$ with $|\lambda| = 1$ and the multiplicative order of λ equals the multiplicative order of u.*

Proof. By Lemma 2.1(3), the characteristic polynomial of L_u on $A_{\mathbb{C}} \cong \mathbb{C}^4$ is $P_u(x)^2$ so its eigenvalues are just the roots of P_u with multiplicity 2. The determinant of a matrix is the constant term of its characteristic polynomial, and the constant term of $P_u(x)^2$ is $\mathrm{Norm}(u)^2$. Since $\mathrm{Norm}(u) = 1$ and $\lambda + \overline{\lambda} = \mathrm{Tr}(u)$, we have that the discriminant of P_u is

$$\mathrm{Tr}(u)^2 - 4\mathrm{Norm}(u) = 4w^2 - 4 = 4(w^2 - 1).$$

Expanding 2.1(1) gives

$$w^2 \leq \mathrm{Norm}(u) = 1$$

which means the discriminant $4(w^2 - 1)$ is nonpositive, while P_u has real coefficients, which means λ, λ' come in conjugate pairs. Now we can write $\lambda\overline{\lambda} = \mathrm{Norm}(u) = 1$ so $\overline{\lambda} = \lambda^{-1} = \lambda'$ and $|\lambda| = |\overline{\lambda}| = 1$.

For the last statement, we note that both u and λ are roots of P_u by Lemma 2.1. So, the fields $\mathbb{Q}(u)$ and $\mathbb{Q}(\lambda)$ are isomorphic. We thus construct the injective (since they are fields so there is no possible nontrivial kernel) ring homomorphism

$$\phi : \mathbb{Q}(u) \to \mathbb{Q}(\lambda)$$

$$a + bu \to a + b\lambda,$$

for $a, b \in \mathbb{Q}$. Since an injective ring homomorphism preserves order, u and λ have the same multiplicative order. □

Left multiplication by $q = (w, x, y, z) \in A_{\mathbb{Q}}$ defines a $\mathbb{Q}$-linear map $A_{\mathbb{Q}} \to A_{\mathbb{Q}}$ with matrix representation

$$L_q \;=\; \begin{bmatrix} w & \alpha x & \beta y & -\alpha\beta z \\ x & w & \beta z & -\beta y \\ y & -\alpha z & w & \alpha x \\ z & -y & x & w \end{bmatrix} \;\in\; \mathfrak{gl}_4(\mathbb{Q}).$$

2.1 The form ψ and its groups

We now use the convention that $a \in A_{\mathbb{Q}}^{\oplus r}$ will be viewed as column vectors

$$a \;=\; \begin{bmatrix} a_1 \\ \vdots \\ a_r \end{bmatrix} \;\in\; A_{\mathbb{Q}}^{\oplus r}$$

with entries $a_i \in A_{\mathbb{Q}}$, such that:

1. Scalar multiplication: let $A_{\mathbb{Q}}$ act on $A_{\mathbb{Q}}^{\oplus r}$ by multiplication on the right.
2. Matrix multiplication: let $GL_r(A_{\mathbb{Q}})$ act on $A_{\mathbb{Q}}^{\oplus r}$ on the left.

This convention ensures that matrix multiplication is $A_{\mathbb{Q}}$–linear.

Definition 2.3 (The form ψ)**.** We define a skew-hermitian quaternionic form

$$\psi : A_{\mathbb{Q}}^{\oplus r} \times A_{\mathbb{Q}}^{\oplus r} \to A_{\mathbb{Q}}$$

as

$$\psi(a,b) = \sum_{i=1}^{r} \overline{a}_i \, \mathbf{I} \, b_{r+1-i}$$

with $a, b \in A_{\mathbb{Q}}^{\oplus r}$, $r \in \mathbb{Z}_{\geq 2}$. For example, if $r = 2$, then $\psi(a,b) = \overline{a}_1 \mathbf{I} b_2 + \overline{a}_2 \mathbf{I} b_1$.

Remark 2.4. *The choice of defining ψ with $\boldsymbol{I}$ here is not forced; as noted later in Remark 3.18, the bound on what characters are possible depends primarily on how ψ is defined. And as also noted in Remark 3.18, a more interesting bilinear form may be*

$$\psi' : \psi'(a,b) = \sum_{i=1}^{r} \overline{a}_i \, (\boldsymbol{I} + \boldsymbol{J} + \boldsymbol{K}) \, b_{r+1-i}$$

which will eventually give the order-6 unit $\frac{1}{2}(1 + \boldsymbol{I} + \boldsymbol{J} + \boldsymbol{K})$. We define ψ as we do above since it was what we initially defined ψ as before finding the result of Theorem 3.16.

We now list some properties of ψ:

Lemma 2.5. *The form ψ is*

1. nondegenerate*: $\psi(a,b) = 0$ for all $b \in A_{\mathbb{Q}}^{\oplus r}$ if and only if $a = 0$;*
2. $A_{\mathbb{Q}}$-linear in the second factor*: $\psi(a, b\,c) = \psi(a,b)c$ for all $c \in A_{\mathbb{Q}}$ and $a, b \in A_{\mathbb{Q}}^{\oplus r}$;*
3. conjugate $A_{\mathbb{Q}}$-linear in the first factor*: $\psi(a\,c, b) = \overline{c}\,\psi(a,b)$ for all $c \in A_{\mathbb{Q}}$ and $a, b \in A_{\mathbb{Q}}^{\oplus r}$;*
4. skew-hermitian*: $\overline{\psi(a,b)} = -\psi(b,a)$.*

Proof. These are all direct consequences of expanding Definition 2.3. □

We can now regard $V_{\mathbb{Q}} = A_{\mathbb{Q}}^{\oplus r} \simeq \mathbb{Q}^{4r}$ as a rational vector space and we define

$$Q_0, Q_1, Q_2, Q_3 : V_{\mathbb{Q}} \times V_{\mathbb{Q}} \to \mathbb{Q}$$

with

$$\psi(a,b) = Q_0(a,b) + Q_1(a,b)\mathbf{I} + Q_2(a,b)\mathbf{J} + Q_3(a,b)\mathbf{K} \tag{2.1.1}$$

Applying Lemma 2.5(4) yields that Q_0 is skew-symmetric, and Q_1, Q_2, Q_3 are symmetric, with the matrices

$$\mathsf{q}_0 = \begin{bmatrix} 0 & \alpha & 0 & 0 \\ -\alpha & 0 & 0 & 0 \\ 0 & 0 & 0 & -\alpha\beta \\ 0 & 0 & \alpha\beta & 0 \end{bmatrix}, \qquad \mathsf{q}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -\alpha & 0 & 0 \\ 0 & 0 & \beta & 0 \\ 0 & 0 & 0 & -\alpha\beta \end{bmatrix},$$

$$\mathsf{q}_2 = \begin{bmatrix} 0 & 0 & 0 & \alpha \\ 0 & 0 & -\alpha & 0 \\ 0 & -\alpha & 0 & 0 \\ \alpha & 0 & 0 & 0 \end{bmatrix}, \mathsf{q}_3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\alpha \\ 1 & 0 & 0 & 0 \\ 0 & -\alpha & 0 & 0 \end{bmatrix}.$$

such that

$$Q_i = \begin{bmatrix} & & \mathfrak{q}_i \\ & \iddots & \\ \mathfrak{q}_i & & \end{bmatrix}.$$

Note that each Q_i is nondegenerate. If we set

$$\Psi = \begin{bmatrix} & & \mathbf{I} \\ & \iddots & \\ \mathbf{I} & & \end{bmatrix} \in \mathfrak{gl}_r(A_{\mathbb{Q}}),$$

then Ψ is the matrix representation of ψ, in the sense that $\psi(a,b) = \overline{a}^T \Psi b$. We can now define

Definition 2.6.

$$SO^*(2r,\mathbb{Q}) = \{g \in GL_r(A_{\mathbb{Q}}) \mid \psi(ga,gb) = \psi(a,b)\ \forall a,b \in A_{\mathbb{Q}}^{\oplus r}\}.$$

Remark 2.7. *By* (2.1.1)*, any $g \in GL_r(A_{\mathbb{Q}})$ preserves ψ iff it preserves each of the four rational forms Q_i. In particular, the action of $SO^*(2r,\mathbb{Q}) \subset GL_r(A_{\mathbb{Q}})$ preserves each Q_i as well.*

Lemma 2.8. *Now we have the identities*

$$\begin{gathered} Q_0(a,b) = Q_1(a,b\boldsymbol{I}) = Q_2(a,b\boldsymbol{J}) = Q_3(a,b\boldsymbol{K}) \\ Q_0(a,b\boldsymbol{I}) = \alpha Q_1(a,b) \\ Q_0(a,b\boldsymbol{J}) = \beta Q_2(a,b) \\ Q_0(a,b\boldsymbol{K}) = -\alpha\beta Q_3(a,b) \end{gathered}$$

Proof. This comes from expanding

$$\psi(a,b\mathbf{I}) = \psi(a,b)\mathbf{I}$$

and similarly for $\mathbf{J}$ and $\mathbf{K}$. □

We now define the Gram matrices with respect to arbitrary bases; these constructions allow us to cleanly capture information at different cusps. We also define orthogonal decompositions of ψ.

Definition 2.9 (Gram matrices; the forms H and $\langle s\rangle$)**.** Let $(b_1,\dots,b_n)$ be an $A_{\mathbb{Q}}$-basis of $V_{\mathbb{Q}}$; i.e. every element of $V_{\mathbb{Q}}$ can be written as $\sum_i b_i c_i$ for some $c_i \in A_{\mathbb{Q}}$. The *Gram matrix* of ψ in this basis is the matrix M such that $M_{ij} = \psi(b_i,b_j) \in A_{\mathbb{Q}}$. By Lemma 2.5 it determines ψ, since $\psi(\sum b_i c_i, \sum b_j d_j) = \sum_{i,j} \overline{c}_i\, \psi(b_i,b_j)\, d_j$. Note that in the standard basis, the gram matrix is Ψ.

We write $\psi \cong \varphi_1 \perp \varphi_2$ if there is a basis in which the Gram matrix is block diagonal with blocks the Gram matrices of φ_1 and φ_2. We now make note of two special forms. Firstly, for a pure quaternion $s \in A_{\mathbb{Q}}$, $\langle s\rangle$ denotes the rank-one skew-hermitian form on $A_{\mathbb{Q}}$ with Gram matrix (s), i.e. $(c_1,c_2) \mapsto \overline{c}_1 s c_2$. Secondly, H is the rank-two form with Gram matrix $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ (i.e. the hyperbolic plane).

2.2 Main setup; period domain, lattice, and arithmetic group

Define

$$A_{\mathbb{C}} = \mathrm{span}_{\mathbb{C}}(1, \mathbf{I}, \mathbf{J}, \mathbf{K}), \qquad V_{\mathbb{C}} = A_{\mathbb{C}}^{\oplus r}$$

Multiplication on the right by $c \in A_{\mathbb{Q}}$ defines a $\mathbb{Q}$-linear map $V_{\mathbb{Q}} \to V_{\mathbb{Q}}$, which extends to a $\mathbb{C}$-linear map, $V_{\mathbb{C}} \to V_{\mathbb{C}}$.

Remark 2.10. *For a $\mathbb{C}$-subspace $E \subseteq V_{\mathbb{C}}$, being preserved by right multiplication by $A_{\mathbb{C}}$ and $A_{\mathbb{Q}}$ are equivalent, so these can be used interchangeably.*

We define $Gr_A^{Q_0}(2r, V_{\mathbb{C}})$ to be the set of all complex subspaces $F \subset V_{\mathbb{C}} \simeq \mathbb{C}^{4r}$ of complex dimension $2r$ that are Q_0-isotropic and preserved under right multiplication by $A_{\mathbb{Q}}$. The group $SO^*(2r, \mathbb{C})$ acts on $Gr_A^{Q_0}(2r, V_{\mathbb{C}})$.

Lemma 2.11. *Let $F \subset V_{\mathbb{C}}$ be any $2r$-dimensional subspace that is isotropic with respect to each of the Q_i, $i = 0, 1, 2, 3$. Then $F \in Gr_A^{Q_0}(2r, V_{\mathbb{C}})$. Conversely, if $F \in Gr_A^{Q_0}(2r, V_{\mathbb{C}})$, then F is isotropic with respect to Q_i, $i = 0, 1, 2, 3$.*

Proof. This is a consequence of Lemma 2.8. □

Remark 2.12 (Explicit low-rank parameterization)**.** *We have computed for $r = 2$ that the isotropic subspaces F of dimension 4 satisfying $F \oplus \overline{F} = V_{\mathbb{C}}$ have a basis parametrized by $t \in \mathbb{C}$, $t \neq \overline{t}$, given by*

$$\begin{aligned} e_1 &= (\sqrt{\alpha}, 1, 0, 0, 0, 0, t\sqrt{\alpha}, -t) \\ e_2 &= (0, 0, \sqrt{\alpha}, 1, \beta t\sqrt{\alpha}, -\beta t, 0, 0) \\ e_3 &= (0, 0, -t\sqrt{\alpha}, t, \sqrt{\alpha}, 1, 0, 0) \\ e_4 &= (-\beta t\sqrt{\alpha}, \beta t, 0, 0, 0, 0, \sqrt{\alpha}, 1). \end{aligned}$$

Definition 2.13 (The period domain D)**.** D is the set of flags $F^1 = V^{1,0} \in Gr_A^{Q_0}(2r, V_{\mathbb{C}})$ satisfying the two Hodge conditions:

1. $F^1 \cap \overline{F^1} = 0$, equivalently $V_{\mathbb{C}} = F^1 \oplus \overline{F^1}$ (so F^1 determines a Hodge decomposition $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$ with $V^{0,1} = \overline{V^{1,0}}$);
2. positivity (second Hodge-Riemann relation): $iQ_0(v, \overline{v}) > 0$ for all $0 \neq v \in F^1$.

Thus D parameterizes weight one, Q_0-polarized Hodge structures on V satisfying the Hodge-Riemann relations as defined above. The group $SO^*(2r, \mathbb{R})$ acts transitively on D.

Remark 2.14 (Low-rank isomorphisms)**.** *For $r = 2$ we have $SO^*(4, \mathbb{R}) \cong (SU(2) \times SL_2(\mathbb{R}))/\{\pm 1\}$ and $D \simeq \mathcal{H}$, the upper half plane, so* $\dim D = 1$. *For $r = 3$ we have $SO^*(6, \mathbb{R}) \cong SU(3, 1)$ up to center, so in this case* $\dim D = 3$.

We can now fix a lattice $V_{\mathbb{Z}} \subset V_{\mathbb{Q}} = A_{\mathbb{Q}}^{\oplus r}$, where we define a lattice as a finitely generated $\mathbb{Z}$-submodule spanning $V_{\mathbb{Q}}$ over $\mathbb{Q}$. The simplest (but not only!) choice of lattice is $A_{\mathbb{Z}}^{\oplus r}$, where $A_{\mathbb{Z}} = \mathrm{span}_{\mathbb{Z}}\{1, \mathbf{I}, \mathbf{J}, \mathbf{K}\}$.

We can now define the arithmetic group Γ:

Definition 2.15 (Γ)**.** If we are working over some lattice $V_{\mathbb{Z}}$, then we can define the arithmetic subgroup Γ for this lattice:

$$\Gamma = \{g \in SO^*(2r, \mathbb{Q}) \mid g(V_{\mathbb{Z}}) = V_{\mathbb{Z}}\}.$$

Happily, we have now defined everything to be able to say that the quotient $\Gamma \backslash D$ is a quasi-projective variety with Satake–Baily–Borel (SBB) compactification $\overline{\Gamma \backslash D}$ [BB66].

2.3 Lie algebra and nilpotent setup

This section is mostly from the overview document; should be the Hodge theoretical motivation .

We have that the Lie algebra of $\mathrm{SO}^*(2r,\mathbb{Q})$ is

$$\mathfrak{so}^*(2r,\mathbb{Q}) = \{X \in \mathfrak{gl}_r(A_\mathbb{Q}) \mid \psi(Xa,b) + \psi(a,Xb) = 0\},$$

where $\mathfrak{gl}_r(A_\mathbb{Q})$ is the set of $r \times r$ matrices with coefficients in $A_\mathbb{Q}$ (not necessarily invertible). Now we define the matrix

$$\mathbf{E} = \begin{bmatrix} & & 1 \\ & \iddots & \\ 1 & & \end{bmatrix} \in \mathfrak{gl}_r(\mathbb{Q}).$$

such that the matrix representation of ψ is $\Psi = \mathbf{EI}$ since $\psi(a,b) = \overline{a}^T \Psi b$. We can now define $\mathfrak{so}^*(2r,\mathbb{Q})$ explicitly, which we write down as the following lemma:

Lemma 2.16. *We have* $\mathfrak{so}^*(2r,\mathbb{Q}) = \{X \in \mathfrak{gl}_r(A_\mathbb{Q}) \mid \overline{X}^T\Psi + \Psi X = 0\}$. *If we express* $X \in \mathfrak{gl}_r(A_\mathbb{Q})$ *as* $X = X_0 + X_1\boldsymbol{I} + X_2\boldsymbol{J} + X_3\boldsymbol{K}$ *for* $X \in \mathfrak{gl}_r(\mathbb{Q})$, *then*

$$\mathfrak{so}^*(2r,\mathbb{Q}) = \{X \in \mathfrak{gl}_r(A_\mathbb{Q}) \mid X_1^T E - EX_1 = 0; X_i^T E + EX_i = 0, i = 0,2,3. \tag{2.3.1}$$

Remark 2.17. *Lemma 2.16 means that* X_1 *is symmetric w.r.t. the diagonal E, and the* $X_i, i = 0,2,3$ *are skew symmetric w.r.t. the diagonal E. In particular,*

$$N = \begin{bmatrix} 0 & 0 \\ \nu & 0 \end{bmatrix} \in \mathfrak{so}^*(4,\mathbb{Q})$$

iff $\nu \in \mathbb{Q}\boldsymbol{I} \subset A_\mathbb{Q}$

Now we state a lemma to connect the nilpotent elements to isotropic lines (which will be the cusps). This connects the isotropic lines to the Hodge structure degeneration and extension behavior.

Lemma 2.18 (Nilpotents attached to isotropic lines)**.** *Pick some nonzero* $v \in V_\mathbb{Q}$ *with* $\psi(v,v) = 0$, *let the the line* $\ell = vA_\mathbb{Q}$, *and let* $0 \neq \lambda \in \mathbb{Q}$. *Now,*

$$N_{v,\lambda}(x) = v\,\lambda\,\psi(v,x)$$

defines a nonzero element of $\mathfrak{so}^*(2r,\mathbb{Q})$ *with*

$$N_{v,\lambda}^2 = 0, \qquad \mathrm{im}(N_{v,\lambda}) = \ell, \qquad \ker(N_{v,\lambda}) = \ell^\perp := \{x \in V_\mathbb{Q} : \psi(v,x) = 0\}.$$

Proof. Using the properties of ψ is stated in Lemma 2.5, and the fact that $\overline{\lambda} = \lambda$ (since $\lambda \in \mathbb{Q}$ is real), and the fact that $\psi(x,v) = -\overline{\psi(v,x)}$ by skew-hermitianness, we write

$$\psi(N_{v,\lambda}x,y) + \psi(x,N_{v,\lambda}y) = \overline{\psi(v,x)}\lambda\,\psi(v,y) + \psi(x,v)\lambda\,\psi(v,y) = 0,$$

to show that N is indeed in $\mathfrak{so}^*(2r,\mathbb{Q})$. Next, $N_{v,\lambda}^2 x = v\lambda\,\psi(v,v)\lambda\,\psi(v,x) = 0$ because v is isotropic. Note that $\lambda\,\psi(v,x) \in A_\mathbb{Q}$ so $N_{v,\lambda}(x) \subseteq \ell$. In addition, the map $x \mapsto \psi(v,x)$ is $A_\mathbb{Q}$-linear with values in $A_\mathbb{Q}$ and nonzero (since ψ is nondegenerate), hence surjective because $A_\mathbb{Q}$ is a division algebra. Therefore, $\mathrm{im}(N_{v,\lambda}) = \ell$ fully, and by definition we thus have $\ker(N_{v,\lambda}) = \ell^\perp$. □

2.4 Boundary components and cusps and the extension character

By [BB66], the boundary components of the SBB compactification $\overline{\Gamma\backslash D}$ are in bijection with the Γ-orbits of nonzero totally ψ-isotropic $A_{\mathbb{Q}}$-subspaces of $V_{\mathbb{Q}}$. (We will see in Lemma 3.5 that for $r = 2, 3$ these subspaces are just $A_{\mathbb{Q}}$-lines.) We are interested in properties of these boundary components; this means that to learn about the boundary components we just need to look at these isotropic subspaces. This motivates a precise definition of the cusps:

Definition 2.19 (Cusps and the standard cusp)**.** A *cusp* in our construction is a nonzero ψ-isotropic $A_{\mathbb{Q}}$-line

$$\ell = vA_{\mathbb{Q}} \subset V_{\mathbb{Q}}, 0 \neq v, \psi(v, v) = 0$$

Two cusps determine the same boundary component iff they are in the same Γ-orbit. The *standard cusp* is $\ell_0 = e_r A_{\mathbb{Q}} = 0 \oplus \cdots \oplus 0 \oplus A_{\mathbb{Q}}$, the line of the last standard basis vector; one computes $\psi(e_r, x) = \mathbf{I}x_1$, so $\ell_0^{\perp} = 0 \oplus A_{\mathbb{Q}}^{\oplus(r-1)}$. $\ell_{\mathbb{C}} = \ell \otimes_{\mathbb{Q}} \mathbb{C}$ is the complexification of ℓ.

Definition 2.20 (Cusp flags)**.** Letting ℓ be a cusp, we attach to it the flag

$$W_{-1} = \ell \subseteq W_0 = \ell^{\perp} \subset V_{\mathbb{Q}}.$$

Note that $\ell \subseteq \ell^{\perp}$ since the cusp is isotropic.

Remark 2.21 (Hodge significance)**.** *By Lemma 2.18 this is exactly the filtration* $\mathrm{im}(N) \subseteq \ker(N)$ *of the nilpotent elements* $N = N_{v,\lambda}$ *attached to* ℓ *(it is the weight filtration of the associated degeneration). For* $r = 3$ *we define the middle of the cusp to be the quotient* $W_0/W_{-1} = \ell^{\perp}/\ell$*, a rank-one right* $A_{\mathbb{Q}}$*-module; for* $r = 2$ *a dimension count (Lemma 3.5) gives* $\ell^{\perp} = \ell$*, so* $W_{-1} = W_0$ *and there is no middle.*

Remark 2.22 (Well-definedness of Q_0 on $\ell^{\perp}/\ell$)**.** *We want* Q_0 *to be well-defined on* $\ell^{\perp}/\ell$ *(so that we can say* Q_0 *extends to* $\ell^{\perp}/\ell$*), which we expand as needing it to be well-defined on cosets* $x + a, y + b$ *for* $x, y \in \ell^{\perp}, a, b \in \ell$*. We expand as requiring that for*

$$Q_0(x + a, y + b) = Q_0(x, y) + Q_0(a, y) + Q_0(x, b) + Q_0(a, b).$$

we want the last three terms to be zero so that $Q_0(x + a, y + b) = Q_0(x, y)$*. Indeed, by the definition of* $\ell^{\perp}$ *and 2.1.1, all the forms* Q_i *vanish on* $\ell \times \ell$ *and* $\ell \times \ell^{\perp}$ *(and vice versa) so* Q_0 *does extend to* $\ell^{\perp}/\ell$*. Similarly, since* ℓ *and* $\ell^{\perp}$ *are defined over* $\mathbb{Q}$ *and complex conjugation doesn't change anything in* $\mathbb{Q}$*, complex conjugation on* $V_{\mathbb{C}}$ *extends to* $(\ell^{\perp}/\ell)_{\mathbb{C}}$*.*

Now we can define the boundary points over a cusp as the possible limits of the Hodge filtrations $F^1 \in D$ as one degenerates toward the cusp. They are described by the data in the following definition; by [GGR] the boundary points over a cusp ℓ are the points of the period subdomain, which we realize as the limiting Hodge filtrations $\pi^{-1}(E)$, and the conditions below are the conditions in 2.13 that survive on $(\ell^{\perp}/\ell)$. A detailed treatment of where these conditions come from can be found in [GGR, §2, §4].

Definition 2.23 (Conditions for boundary points over a cusp)**.** Choose any cusp ℓ and let $\pi : \ell_{\mathbb{C}}^{\perp} \to (\ell^{\perp}/\ell)_{\mathbb{C}}$ be the quotient map (note that π is the complexification of the rational map $\pi_{\mathbb{Q}} : \ell^{\perp} \to (\ell^{\perp}/\ell)$ so π is a rational map as well).

1. For $r = 2$, the unique boundary point over ℓ is $F_{\infty} = \ell_{\mathbb{C}}$, of complex dimension $4 = 2r$ (recall Remark 2.21: $\dim_{\mathbb{C}} \ell_{\mathbb{C}} = 4$ so there is no space for more interesting cusps).

2. For $r = 3$: a boundary point over ℓ is a subspace with internal decomposition

$$F_\infty \;=\; \pi^{-1}(E) \;\subset\; \ell_{\mathbb{C}}^{\perp},$$

of complex dimension $6 = 2r$. where $E \subset (\ell^{\perp}/\ell)_{\mathbb{C}}$ is a complex 2-plane satisfying:

(a) E is preserved by right multiplication by $A_{\mathbb{Q}}$;

(b) $E \cap \overline{E} = 0$;

(c) *positivity*: $iQ_0(e, \overline{e}) > 0$ for all $0 \neq e \in E$ (well defined by the preceding remark).

(d) $Q_0(e, e') = 0$ for all $e, e' \in E$.

We will see in Lemma 3.7 that we can construct a basis of $A_{\mathbb{C}}^{\oplus 3}$ such that we can write this flag explicitly in the familiar form

$$F_\infty = \pi^{-1}(E) = \{x \in \ell_{\mathbb{C}}^{\perp} : \pi(x) \in E\} = 0 \oplus \tilde{E} \oplus \ell_{\mathbb{C}} \subset A_{\mathbb{C}}^{\oplus 3}$$

where $\tilde{E}$ is any lift of E from $(\ell^{\perp}/\ell)_{\mathbb{C}}$ to $\ell_{\mathbb{C}}^{\perp}$.

Note that $F_\infty \supseteq \ell_{\mathbb{C}}$ always, and that F_∞ is preserved by right multiplication by $A_{\mathbb{Q}}$.

Lemma 2.24 (Stabilizing a boundary point stabilizes the flag)**.** *Let some $g \in \Gamma$ stabilize a boundary point F_∞ over a cusp ℓ; i.e. $g(F_\infty) = F_\infty$. Thus g preserves the whole flag in Definition 2.20; i.e. $g(\ell) = \ell$ and $g(\ell^{\perp}) = \ell^{\perp}$*

Proof. For $r = 2$ this is immediate since $F_\infty = \ell_{\mathbb{C}}$ with no weird things that need to be preserved. For $r = 3$, we have that since g is rational (thus real). $g(F_\infty) = F_\infty \implies g(\overline{F_\infty}) = \overline{F_\infty}$, and hence g will also stabilize the intersection $F_\infty \cap \overline{F_\infty}$. Furthermore, since π is a rational map (i.e. real), π commutes with conjugation, giving us $\overline{F_\infty} = \pi^{-1}(\overline{E})$. We can thus write that g preserves

$$F_\infty \cap \overline{F_\infty} = \pi^{-1}(E \cap \overline{E}) = \pi^{-1}(0) = \ell_{\mathbb{C}}.$$

and g preserves $\ell_{\mathbb{C}}$ means it preserves $\ell = \ell_{\mathbb{C}} \cap V_{\mathbb{Q}}$ as well. Now since g is an isometry of ψ we have $g(\ell^{\perp}) = \ell^{\perp}$ as well. □

Definition 2.25 (Γ_∞ and the extension character χ_∞)**.** Fix a cusp ℓ and a boundary point F_∞ over it. Define

$$\Gamma_\infty \;=\; \{g \in \Gamma \mid g(F_\infty) = F_\infty\},$$

the stabilizer of the boundary point (which preserves the flag of ℓ by Lemma 2.24). For this choice of ℓ, the *extension character* is

$$\chi_\infty : \Gamma_\infty \to \mathbb{C}^\times, \qquad \chi_\infty(g) \;=\; \det_{\mathbb{C}}\big(g|_{F_\infty}\big).$$

Theorem 2.26 (GGR)**.** *The values of χ_∞ are roots of unity. The n-th power $\det(F^1)^{\otimes n}$ of the Hodge line bundle $\det(F^1) \to \Gamma\backslash D$ extends to the SBB compactification $\overline{\Gamma\backslash D}$ if and only if $\chi_\infty^n \equiv 1$ for every boundary point F_∞ over every cusp ℓ. We call the smallest such $n \geq 1$ the* extension power *of the construction; it is the least common multiple, over all boundary points of all boundary components, of the orders of the values of χ_∞.*

Our goal is for any choice of $\alpha, \beta \in \mathbb{Q}_{<0}$ and every choice of lattice $V_{\mathbb{Z}}$, to bound the extension character over all boundary components, which will thus bound the extension power.

3 Bounding the extension power

Definition 3.1 (Full lattices)**.** A *full $\mathbb{Z}$-lattice* $L \subset A_{\mathbb{Q}}$ is a finitely generated $\mathbb{Z}$-submodule containing a $\mathbb{Q}$-basis of $A_{\mathbb{Q}}$; i.e. it spans all of the 4-dimensional $A_{\mathbb{Q}}$ over $\mathbb{Q}$.

Definition 3.2. Let $L \subset A$ be any full $\mathbb{Z}$-lattice; orders in A are thus full subrings of L. In particular, we define $S_L = \{x \in A : xL \subseteq L\}$ to be the left multiplier ring of L (which, importantly, is closed under this multiplication operation); S_L is now an order in $A_{\mathbb{Q}}$ [Voi21, Ch. 10].

Remark 3.3 (Why we care about orders)**.** *An order S_L on a lattice L is the lattice analogue of the ring of integers $\mathcal{O}_K$ on a number field K; in particular, it gives the order S_L a unit group. We want this because an element $g \in \Gamma_\infty$ will act on each rank-one piece of the flag of its cusp by left multiplication by some quaternion u, and the lattice preservation that we need, $g(V_{\mathbb{Z}}) = V_{\mathbb{Z}}$, forces u to map the corresponding sublattice L bijectively onto itself. Thus u and u^{-1} both lie in S_L, i.e. u is a* unit *of the order S_L.*

Lemma 3.4 (Units on orders)**.** *Let S be any order in $A_{\mathbb{Q}}$ and $u \in S^\times$ be a unit with $u^{-1} \in S$. Then* $\mathrm{Norm}(u) = 1$.

Proof. Let S be an order. By Definition 3.2, S is a subring that is a finitely generated $\mathbb{Z}$-module, so S is integral over $\mathbb{Z}$. Therefore for any element $x \in S, \mathrm{Norm}(x)$ is also integral (since x integral implies $\overline{x} = \mathrm{Tr}(x) - x$ integral, and $\mathrm{Norm}(x) = x\overline{x}$) and in $\mathbb{Z}$ (since the norm is an integral rational). And because units are invertible, for some unit u in the order we have that $\mathrm{Norm}(u)$ and $\mathrm{Norm}(u^{-1}) = \mathrm{Norm}(u)^{-1}$ are are both positive (by Lemma 2.1) integers, so $\mathrm{Norm}(u) = 1$. □

Lemma 3.5 (Isotropic subspaces are lines for $r = 2, 3$)**.** *Every totally ψ-isotropic A-subspace of $A^{\oplus r}$ has A-dimension at most $\lfloor r/2 \rfloor$. In particular, for $r = 2, 3$ the rational boundary components of $\Gamma\backslash D$ correspond to Γ-orbits of cusps $\ell = vA_{\mathbb{Q}} \subset V_{\mathbb{Q}}$ (as defined in Definition 2.19), and there are no deeper rational flags. Moreover for $r = 2$, every cusp satisfies $\ell^\perp = \ell$.*

Proof. For some totally isotropic E of dimension k we have that $E \subseteq E^\perp$. Since ψ is nondegenerate, we now have that $\dim_A E^\perp + \dim_A E = \dim_A V \implies \dim_A E^\perp = r - k$. These two combined mean that $k \leq r - k$ which gives us us the $\lfloor r/2 \rfloor$ bound. The second statement now follows from the correspondence between boundary components and Γ-orbits of totally isotropic subspaces [BB66], since for $r = 2, 3$ the only such subspaces are lines. □

Remark 3.6 (Motivation for next step)**.** *For the standard cusp, the flag in Definition 2.20 was aligned with our coordinate system. Explicitly, the cusp in standard coordinates was already $0 \oplus 0 \oplus A_{\mathbb{Q}}$; W_{-1} is the last coordinate and W_0 is the last $r - 1$ coordinates. So, to preserve the flag we just needed that any $g \in \Gamma_\infty$ is lower triangular, and expanding $\overline{g}^T \Psi g = \Psi$ gives usable entry-by-entry conditions. But for some new cusp $\ell = vA_{\mathbb{Q}}$, ℓ may not be a coordinate line in $V_{\mathbb{Q}}$, in which case the flag F_∞ will not align with the coordinates of $V_{\mathbb{Q}}$, which means that the action by $g \in \Gamma_\infty$ will no longer be triangular in the standard coordinates. So, we build a basis w, m, v aligned to the new flag; over $A_{\mathbb{Q}}$, we have v spanning the cusp ℓ, corresponding to last coordinate, a partner coordinate w that isotropic and pairs invertibly with v corresponding to the first coordinate, and m spanning the orthogonal complement of the pair, representing the middle W_0/W_{-1}. This gives us the matrix Ψ' below which is the Gram matrix of ψ as defined in Definition 2.9 in the new basis; we choose the basis to force that Ψ' remains antidiagonal, allowing us to carry out the character computation on new cusps.*

Lemma 3.7 (Adapted cusp basis)**.** *Let $\ell = vA_{\mathbb{Q}}$ be an isotropic line on the $r = 3$ construction. There is now an $A_{\mathbb{Q}}$-basis (w, m, v) of $V_{\mathbb{Q}}$, with Gram matrix (Definition 2.9)*

$$\Psi' = \begin{bmatrix} 0 & 0 & \tau \\ 0 & s & 0 \\ -\overline{\tau} & 0 & 0 \end{bmatrix} : \tau = \psi(w, v), s = \psi(m, m) \tag{3.0.1}$$

Proof. We take some $w_0 : \psi(w_0, v) \neq 0$, which we can do because ψ is nondegenerate. Now set $w = w_0 + vd$ for some $d \in A_{\mathbb{Q}}$; we can now correct this to be isotropic. To construct the isotropy, take

$$\psi(w, w) = \psi(w_0, w_0) + \overline{d}\psi(v, w_0) + \psi(w_0, v)d + \overline{d}\psi(v, v)d = \psi(w_0, w_0) + (\tau_0 d - \overline{\tau_0 d})$$

for $\tau_0 = \psi(w_0, v)$. Now, $(\tau_0 d - \overline{\tau_0 d})$ is twice the pure part of $\tau_0 d$, and $\psi(w_0, w_0)$ is pure by skew-hermitianness of ψ. Therefore, we can therefore now set the isotropy condition and solve for d:

$$\begin{aligned} \psi(w, w) = \psi(w_0, w_0) + (\tau_0 d - \overline{\tau_0 d}) = 0 \\ \implies 2\text{pure}(\tau_0 d) = -\psi(w_0, w_0). \end{aligned}$$

Picking a d that satisfies the above equation therefore gives us a working $w = w_0 + dv$. We can now take some m spanning the orthogonal ψ-complement of $vA_{\mathbb{Q}} \oplus wA_{\mathbb{Q}}$, completing our adapted basis. □

Remark 3.8. *At the standard cusp, $\Psi' = \Psi = \begin{bmatrix} & & \boldsymbol{I} \\ & \boldsymbol{I} & \\ \boldsymbol{I} & & \end{bmatrix}$.*

Remark 3.9. *Since ψ is skew-hermitian we deduce that $\overline{s} = -s \neq 0$ and therefore s is a pure quaternion.*

Lemma 3.10 (Conditions on the parabolic Γ_∞ at general cusps)**.** *Let $\ell = vA_{\mathbb{Q}}$ be a cusp of the $r = 3$ construction with flag $W_{-1} = \ell \subset W_0 = \ell^\perp$ (Definition 2.20), and fix an adapted basis (w, m, v) as in Lemma 3.7, so that in the coordinates $x = wc_1 + mc_2 + vc_3$ we have $W_{-1} = \{c_1 = c_2 = 0\}$ and $W_0 = \{c_1 = 0\}$. Let $F_\infty = \pi^{-1}(E)$ be a boundary point over ℓ (with E the complex 2-plane as defined in Definition 2.23), and let $g \in \Gamma_\infty$. g is therefore lower triangular in this basis, and we can expand it as*

$$g = \begin{bmatrix} g_{11} & & \\ g_{21} & g_{22} & \\ g_{31} & g_{32} & g_{33} \end{bmatrix}$$

for $g_{ij} \in A_{\mathbb{Q}}$. Now, we have the following conditions:

1. *$\overline{g}_{22} s g_{22} = s$ and $\overline{g}_{11}\, \tau\, g_{33} = \tau$;*
2. *g_{33}, g_{33}^{-1} both lie in S_I where $I = \{c \in A_{\mathbb{Q}} : vc \in V_{\mathbb{Z}}\}$ and S_I denotes the order over I given by Definition 3.2. Similarly, g_{22} and g_{22}^{-1} are in S_M, where S_M is the order over $M = \{c \in A_{\mathbb{Q}} : mc \in (W_0 \cap V_{\mathbb{Z}}) + \ell_{\mathbb{Q}}\}$. Both I, M are full lattices in $A_{\mathbb{Q}}$. Conceptually, S_M is the order on the middle lattice, written as the intersection of the flag with the lattice, in terms of the adapted coordinate m, and S_I is the order on the last lattice.*

3. for g preserving a boundary point $F_\infty = E \oplus W_{-1,\mathbb{C}}$ over this cusp,

$$\chi_\infty(g) = \det_{\mathbb{C}}(L_{g_{22}|_E})\det_{\mathbb{C}}(L_{g_{33}})$$

Proof. Lower triangularity comes from g preserving the whole flag. We can expand

$$\begin{bmatrix} \overline{g}_{11} & \overline{g}_{21} & \overline{g}_{31} \\ 0 & \overline{g}_{22} & \overline{g}_{32} \\ 0 & 0 & \overline{g}_{33} \end{bmatrix} \begin{bmatrix} 0 & 0 & \tau \\ 0 & s & 0 \\ -\overline{\tau} & 0 & 0 \end{bmatrix} \begin{bmatrix} g_{11} & 0 & 0 \\ g_{21} & g_{22} & 0 \\ g_{31} & g_{32} & g_{33} \end{bmatrix} = \begin{bmatrix} 0 & 0 & \tau \\ 0 & s & 0 \\ -\overline{\tau} & 0 & 0 \end{bmatrix}. \tag{3.0.2}$$

Now, matching terms gives us (1) directly. For (2), we first have that from the triangular form, $g(vc) = v(g_{33}c)$ and $g(mc) \equiv m(g_{22}c) \bmod \ell_{\mathbb{Q}}$. Now, both $g^{-1} = \begin{bmatrix} g_{11}^{-1} & 0 & 0 \\ * & g_{22}^{-1} & 0 \\ * & * & g_{33}^{-1} \end{bmatrix}$ and g must preserve $V_{\mathbb{Z}}$ (i.e. $gV_{\mathbb{Z}} = V_{\mathbb{Z}}$) and the whole flag, and hence the respective sublattices I and M, i.e. $g_{33}, g_{33}^{-1} \in S_I$ and $g_{22}, g_{22}^{-1} \in S_M$. For (3), the flag we want to preserve F_∞ has a basis consisting of lifts $\tilde{e}_1 = m\,\varepsilon_1, \tilde{e}_2 = m\,\varepsilon_2$ (where $\varepsilon_1, \varepsilon_2 \in A_{\mathbb{C}}$ span E in the $[m]$-coordinate, and the lift is from $(\ell^\perp/\ell)_{\mathbb{C}} \to \ell_{\mathbb{C}}^\perp$ as seen in Definition 2.23). In this basis the action of g on F_∞ is the block lower triangular matrix

$$g|_{F_\infty} \;=\; \begin{bmatrix} L_{g_{22}}|_E & 0 \\ * & L_{g_{33}} \end{bmatrix},$$

where $L_u : A_{\mathbb{C}} \to A_{\mathbb{C}}$ represents left multiplication by u as used in Lemma 2.2 since $g(m\varepsilon) = m(g_{22}\varepsilon) + v(g_{32}\varepsilon)$ (so the only thing sending an m-coordinate back to an m-coordinate is the left action of g_{22}) and $g(vc) = v(g_{33}c)$. In particular $g(F_\infty) = F_\infty$ if and only if $g_{22}E = E$, and the overall matrix determinant is the product of the two diagonal block determinants. □

Remark 3.11. *Note that g_{11} acts only on the w-coordinate of our adapted basis; however, the flag of our cusp F_∞ lies in $\ell_{\mathbb{C}}^\perp = W_0$, and we have adapted our basis to this cusp such that the w-component of F_∞ is zero. Thus g_{11} does not show up in the final character; only g_{22} and g_{33} do.*

We now have enough lemmas to prove the simple case where $r = 2$; after this we will finish the $r = 3$ proof (all notation for the theorem below is just for this $r = 2$ case; after this treat g, ℓ, etc. as before.):

Theorem 3.12 (Extension on $r = 2$)**.** *For $r = 2$, on every lattice $V_{\mathbb{Z}}$ on every $\alpha, \beta \in \mathbb{Q}_{<0}$ and every boundary component, $\chi_\infty = 1$. Hence $\det(F^1)$ extends to $\overline{\Gamma\backslash D}$ on the nose.*

Proof. Here we have by Lemma 3.5 that $\ell^\perp = \ell$ with no middle component, and by Definition 2.23, the boundary point over this cusp is just $F_\infty = \ell_{\mathbb{C}} \cong A_{\mathbb{C}}$. The analogue of Lemmas 3.7 and 3.10 gives an adapted basis (w, v) with gram matrix in the basis of $\Psi' = \begin{bmatrix} 0 & \tau \\ -\overline{\tau} & 0 \end{bmatrix}$ and lower triangular $g = \begin{bmatrix} g_{11} & \\ g_{21} & g_{22} \end{bmatrix}$ gives that any $g \in \Gamma_\infty$ acts on the cusp $\ell_{\mathbb{C}}$ by left multiplication by a unit u in the order S_I, where $I = \{c : vc \in V_{\mathbb{Z}}\}$. The character is now just the determinant of the left action on ℓ. So, by Lemmas 3.4 and 2.2, we have

$$\chi_\infty(g) = \det_{\mathbb{C}}(L_u) = \mathrm{Norm}(u)^2 = 1.$$

□

Now for $r = 3$, by Lemma 3.10(3) and Lemma 2.2 the extension power just depends on what $u = g_{22}$ can be since $g_{33} \in S_I$, giving $\det_{\mathbb{C}}(g_{33}) = \text{Norm}(g_{33})^2 = 1$.

Lemma 3.13. *[Conditions from the fact that the middle entry commutes with s] Let $u = g_{22}$ as defined above. We now have*

$$u = p + qs, \qquad p, q \in \mathbb{Q}.$$

Proof. By Lemma 3.10(2) we have $\text{Norm}(u) = u\overline{u} = 1 \implies \overline{u} = u^{-1}$. Lemma 3.10(1) gives us that $us\overline{u} = s$, and applying the previous statement turns this into gives us that $us = su$. Let $u = p + u'$ for some pure u'; since a scalar p commutes with everything, we get

$$ps + u's = sp + su' \implies u's = su'$$

We now claim that a pure quaternion that commutes with s (which recall is pure by skew-hermitianness of ψ) must be a rational multiple of s. This is because for pure u', s, $\overline{u's} = \overline{s}\,\overline{u'} = su'$ by Lemma 2.1, so

$$u's + su' = u's + \overline{u's} = \text{Tr}(u's) \in \mathbb{Q}$$

combined with $u's = su'$ gives $2u's = \text{Tr}(u's) \implies u' = \frac{1}{2}\text{Tr}(u's)s^{-1} \in \mathbb{Q}s$ since $s^{-1} = -s/\text{Norm}(s) \in \mathbb{Q}s$. Thus $u' = qs$ for some rational q and $u = p + qs$.

□

Lemma 3.14 (Torsion)**.** *With some $u = p + qs$ as defined above, u is a root of unity of order $1, 2, 3, 4$ or 6, such that*

1. *Order 4 only occurs if $q^2s^2 = -1$*
2. *Order 3 or 6 only occurs if $q^2s^2 = -\frac{3}{4}$*

Proof. By Lemma 2.1(3), the characteristic polynomial of the action of left multiplication by u over $\mathbb{Q}$, L_u, is $(x^2 - 2px + \text{Norm}(u))^2 = (x^2 - 2px + 1)^2$. Since u maps the lattice M bijectively to itself, the $\mathbb{Q}$-matrix of L_u, in some $\mathbb{Z}$-basis of M, must be an integer matrix. This gives us that $2p \in \mathbb{Z}$ since an integer matrix will have a polynomial with integer constraints. Also recall that for pure quaternions s, $-\text{Norm}(s) = s^2 < 0$ (anticommutativity cancels the cross terms). We thus get

$$1 = \text{Norm}(u) = p^2 - q^2s^2 \geq p^2.$$

This confines $2p \in \{0, \pm 1, \pm 2\}$. Now we perform casework:

1. If $2p = 0$, then we get $u^2 = -1$ (order 4), with $q^2s^2 = -1$ from the norm equation.
2. If $2p = \pm 1$, then we get $u^2 \mp u + 1 = 0$ (orders 6 and 3), with $q^2s^2 = -\frac{3}{4}$ from the norm equation
3. If $2p = \pm 2$, then we get $u^2 + 2u + 1 = 0$ (orders 2 and 1), with $q^2s^2 = 0$ from the norm equation.

□

Lemma 3.15 (All possible s from Witt cancellation)**.** *For any cusp ℓ of the $r = 3$ construction on $A_{\mathbb{Q}}^{\oplus 3}$, with $s = \psi(m, m)$ (i.e. the middle gram element from Lemma 3.7), we have*

$$s^2 = \alpha \cdot \mathrm{Norm}(c)^2 \qquad \textit{for some } c \in A_{\mathbb{Q}}^{\times}.$$

In particular, at every cusp, the squarefree part of $s^2 \in \mathbb{Q}_{<0}$, and therefore the quadratic field $\mathbb{Q}(s) \cong \mathbb{Q}(\sqrt{\alpha})$, is the same.

Proof. In the notation of Definition 2.9, Lemma 3.7 exhibits the block diagonal splitting

$$\psi = H \perp \langle s \rangle : \tag{3.0.3}$$

Here, H is isometric to the hyperbolic plane spanned by v, w. One can see this by sending $w \to w\overline{\tau}^{-1}$, after which the Gram matrix for ψ on the plane $wA_{\mathbb{Q}} \oplus vA_{\mathbb{Q}}$ becomes $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, which is the hyperbolic plane H. By Lemma 2.5(3) this is still isotropic. The middle line contributes the block $\langle s \rangle$. All such planes are isometric because for any v, w, by Lemma 2.1, $A_{\mathbb{Q}}$ is a division algebra; $\tau = \psi(w, v) \in A_{\mathbb{Q}}$ is thus invertible. Applying the same construction at the standard cusp, where $\tau = s = \mathbf{I}$ by Remark 3.8, yields that in particular,

$$\psi = H \perp \langle \mathbf{I} \rangle. \tag{3.0.4}$$

By Witt cancellation for hermitian forms over a division algebra with involution, we can now cancel H from Equations 3.0.3 and 3.0.4 ([Sch85] Ch.7). This yields that

$$\langle s \rangle = \langle \mathbf{I} \rangle$$

over $\mathbb{Q}$. Since an isometry of rank-one forms (which $\langle s \rangle, \langle \mathbf{I} \rangle$ are) is just a change of generator, i.e. left multiplication by an invertible quaternion, we have that for any s, we have that $s = \overline{c}\mathbf{I}c$ for some $c \in A_{\mathbb{Q}}^{\times}$. Recalling that $\overline{c}c = \mathrm{Norm}(c)$, we now have

$$s^2 = \overline{c}\mathbf{I}c\overline{c}\mathbf{I}c = \mathrm{Norm(c)}\overline{c}\mathbf{I}^2 c = \alpha \mathrm{Norm}(c)^2.$$

Lastly, this is well defined up to the squarefree part of s^2 because for some arbitrary $h \in A_{\mathbb{Q}}^{\times}$ can change basis to $\overline{h}sh$, which will multiply s^2 by $\mathrm{Norm}(h)^2$. □

Theorem 3.16 (Extension power bounds for $r = 3$)**.** *Let $r = 3$, let $\alpha, \beta \in \mathbb{Q}_{<0}$ be arbitrary, let $V_{\mathbb{Z}} \subset A_{\mathbb{Q}}^{\oplus 3}$ be an arbitrary lattice, and let d be the (positive) squarefree part of α. Then at every boundary component of $\overline{\Gamma \backslash D}$,*

$$\chi_\infty \in \begin{cases} \{1\}, & d \neq 1, 3, \\ \{\pm 1\}, & d = 1, \\ \{1, \zeta_3, \zeta_3^2\}, & d = 3. \end{cases}$$

The extension power thus divides 1, 2, or 3, respectively. It also is independent of β and independent of any square factor of α.

Proof. By Lemmas 3.10(3) and 2.2, we have $\chi_\infty(g) = \det_{\mathbb{C}}(u|_E) \cdot \mathrm{Norm}(g_{33})^2 = \det_{\mathbb{C}}(u|_E)$ where by Lemma 3.10(2), $u = g_{22}$. We now combine Lemmas 3.14 and 3.15 in casework, where $\alpha = -dt^2$ with $t \in \mathbb{Q}^{\times}$:

1. For u to be order 4, we want $q^2s^2 = \alpha(\mathrm{Norm}(c)q)^2 = -1$, i.e. $d(t\mathrm{Norm}(c)q)^2 = 1$. We thus need $d = 1$ since a positive squarefree integer d times a rational square can only equal 1 if $d = 1$.

2. For u to be order 3 or 6, we want $q^2s^2 = \alpha(\mathrm{Norm}(c)q)^2 = -\frac{3}{4}$. For analogous reasons as above, this requires $d = 3$.

3. Otherwise, $u = \pm 1$.

Lemma 2.2 gives us that the eigenvalues of the left action L_u of a a unit u on $A_{\mathbb{C}}$ are some roots of unity $\lambda, \overline{\lambda}$ with multiplicity 2. The order of an eigenvalue λ is the same as the order of u by Lemma 2.2. The E-plane is u-invariant by Lemma 3.10(3), so the characteristic polynomial of $u|_E$ divides the characteristic polynomial of L_u (take e_1, e_2 a basis of E and extend this to a basis e_1, e_2, f_1, f_2 of $A_{\mathbb{C}}$; now the action of $u|_E$ is block triangular in this basis so the characteristic polynomials divide). Therefore, the two eigenvalues of $u|_E$ must be in the set $\{\lambda, \overline{\lambda}\}$. We thus have

$$\det_{\mathbb{C}}(u|_E) \in \{\lambda^2, \lambda\overline{\lambda}, \overline{\lambda}^2\}.$$

For $u = \pm 1$ this is just $\{1\}$. For some order 4 u, this is $\{\pm 1\}$. For some order 3 or 6 u, this is $\{1, \zeta_3, \zeta_3^2\}$ (squaring turns order 6 into order 3). □

Remark 3.17 (Role of the cusp classification)**.** *The only cusp-dependent data that enters the proof are the order S_M and the Gram element s, and Lemma 3.15 controls s uniformly. So the bound requires no classification of the Γ-conjugacy classes of isotropic sublattices; the classification matters only for whether the bound is attained, i.e. whether at some cusp the torsion unit $u = p + qs$ of order 4 (resp. 6) lies in S_M and extends to an element of Γ_∞. Different β (and different square factors of α) change the algebra and the lattice, so attainment is a separate finite check for each construction. Fortunately now we only need to find a single good cusp, instead of enumerating all of them.*

Remark 3.18 (Asymmetry between α and β)**.** *The asymmetry between α and β is an artifact of how we defined our bilinear form ψ; as seen in Lemma 3.15, the quadratic field that controls our character is $\mathbb{Q}[s] \cong \mathbb{Q}[\boldsymbol{I}] = \mathbb{Q}(\sqrt{\alpha})$. If we built ψ from some other quaternion s_0 (instead of $\boldsymbol{I}$), then we would have the squarefree part of s_0^2 would play the role of the squarefree part of α. For example, if we built our quaternion field with $\alpha = \beta = -1$ and used $s_0 = \boldsymbol{I} + \boldsymbol{J} + \boldsymbol{K}$ to define ψ instead of $s_0 = \boldsymbol{I}$, the order-6 unit $\frac{1}{2}(1 + \boldsymbol{I} + \boldsymbol{J} + \boldsymbol{K})$ would become available.*

Remark 3.19 (Non-integer α, β)**.** *Since only the squarefree part of α matters, any α will behave like $\frac{1}{\alpha}$; we can always choose $(\mathrm{Norm}(c)q)^2$ to cancel any square part of alpha, and the difference between α and $\frac{1}{\alpha}$ is only the square α^2.*

3.1 Examples of attainment of the bounds

Recall that by 3.17 we have only given bounds for what the the extension power can be, but for $d = 1, 3$ each separate construction needs a separate check to exhibit the higher-order behavior. It suffices to exhibit just a single element preserving the filtration with higher-order behavior. Here we provide examples of this higher-order behavior.

Example 3.20 ($d = 1$; order two attainment)**.** *For $\alpha = \beta = -1$, the squarefree part of α is $d = 1$, so Theorem 3.16 gives a extension power bound of 2. On this construction, the standard cusp where $g = \mathrm{diag}(1, \boldsymbol{I}, 1)$ lies in Γ (since the conditions in Lemma 3.10 hold). Of the two*

eigenplanes of $L_{\boldsymbol{I}}$ on the middle $A_{\mathbb{C}}$, one is positive definite (which we need by Definition 2.23), and on it $\chi_\infty(g) = \det_{\mathbb{C}}(\boldsymbol{I}_E) = -1$, giving us a lower bound for the extension power of 2, which matches our paper's upper bound of 2 for $d = 1$.

Definition 3.21 (Lipschitz-type and ω-order lattices)**.** Fix $(\alpha, \beta) = (-3, -1)$. Since $\mathbf{I}^2 = -3$, the element $\omega = \frac{1}{2}(1 + \mathbf{I})$ satisfies $\omega^2 - \omega + 1 = 0$: it is an integral primitive sixth root of unity (the case $p = q = \frac{1}{2}$, $s = \mathbf{I}$ of Lemma 3.14). We call $A_{\mathbb{Z}} = \mathbb{Z}\langle 1, \mathbf{I}, \mathbf{J}, \mathbf{K}\rangle$ the *Lipschitz-type lattice*, and

$$L' \;=\; \mathbb{Z}\langle 1,\ \omega,\ \mathbf{J},\ \omega\mathbf{J}\rangle$$

the *ω-order lattice*. Expanding products shows that L' is closed under multiplication; it is an order containing $A_{\mathbb{Z}}$ with index 4, of reduced discriminant 3 (hence maximal), while $A_{\mathbb{Z}}$ has reduced discriminant 12. This is the $\alpha = -3$ analogue of the Lipschitz order inside the Hurwitz order: the maximal order is obtained by adjoining the half-integral root of unity.

Example 3.22 ($d = 3$; order three attainment on ω-order lattice)**.** *Let $(\alpha, \beta) = (-3, -1)$ and $V'_{\mathbb{Z}} = L'^{\oplus 3}$. Then the extension power is exactly 3.*

Proof. By Theorem 3.16 the extension power divides 3, so we need to exhibit an element realizing ζ_3. Let $g = \mathrm{diag}(\omega, \omega, \omega)$. Since $\omega L' = L'$, we have that g preserves $V'_{\mathbb{Z}}$. Furthermore, ω preserves ψ because $\omega \in \mathbb{Q}(\mathbf{I})$ commutes with $\mathbf{I}$ giving $\overline{\omega}\mathbf{I}\omega = \mathrm{Norm}(\omega)\mathbf{I} = \mathbf{I}$. g acts on the middle by $g_{22} = u = \omega$ at the standard cusp. Of the two eigenplanes of ω on $A_{\mathbb{C}}$, only the $e^{i\pi/3}$-eigenplane is positive definite, so we must choose that as E. This gives us

$$\chi_\infty(g) = \det_{\mathbb{C}}(\omega|_E) \cdot \mathrm{Norm}(\omega)^2 = (e^{i\pi/3})^2 = \zeta_3.$$

which indeed exhibits a third root of unity needed to extend. □

Example 3.23 ($d = 3$; order three attainment on Lipschitz-type lattice)**.** *For $V_{\mathbb{Z}} = A_{\mathbb{Z}}^{\oplus 3}$ with $(\alpha, \beta) = (-3, -1)$, the bound 3 of Theorem 3.16 holds, and an exhaustive exact-arithmetic computer search (scanning the primitive isotropic vectors v in the coordinate box $\{-1, 0, 1\}^{12}$, and for each computing the middle lattice M and Gram element s and testing whether the order-six unit $u = \frac{1}{2}(1 \pm s/k)$, $k^2 = -s^2/3$, preserves M (thanks Fable)) finds many cusps carrying order-six torsion:* 192 *primitive vectors, lying on* 48 *distinct lines, whose middle lattice is preserved by an order-six unit with valid positive eigenplane E; at the first of them, $s = 4\boldsymbol{I}$ and $u = \frac{1}{2}(1 + \boldsymbol{I}) \in S_M$. (At the* standard *cusp $M = A_{\mathbb{Z}}$ and $\omega \notin S_M$, so the order-six torsion appears only at non-standard cusps: inequivalent isotropic sublattices carry different boundary behavior here.) It is still open for this lattice whether such a torsion element lifts to Γ_∞, i.e. whether the entries $g_{11}, g_{21}, g_{31}, g_{32}, g_{33}$ can be filled in integrally; the script was not enough to see which one it is. So for $A_{\mathbb{Z}}^{\oplus 3}$ the extension power is* 1 *or* 3*, while for $L'^{\oplus 3}$ it is exactly* 3.

References

[BB66] W. L. Baily Jr. and A. Borel, *Compactification of arithmetic quotients of bounded symmetric domains*, Ann. of Math. (2) **84** (1966), 442–528.

[Sch85] W. Scharlau, *Quadratic and Hermitian Forms*, Grundlehren der mathematischen Wissenschaften, vol. 270, Springer-Verlag, Berlin, 1985.

[GGR] M. Green, P. Griffiths, and C. Robles, *Analog of Satake–Baily–Borel for period maps.* arXiv preprint (2025).

[Voi21] J. Voight, *Quaternion Algebras*, Graduate Texts in Mathematics, vol. 288, Springer International Publishing, 2021.